

PLAN OVERVIEW

A Data Management Plan created using DMPTool

Title: 1Florida Alzheimer's Disease Research Center

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1FLORIDA ALZHEIMER'S DISEASE RESEARCH CENTER

DATA TYPE

ELEMENT 1: TYPES AND AMOUNT OF SCIENTIFIC DATA EXPECTED TO BE GENERATED IN THE PROJECT

We use human subjects data including information from, to date, 1,009 adult registrants of the 1Florida Alzheimer's Disease Research Center (1Florida). By the conclusion of this five-year competing continuation, we expect to have at least one occasion of measurement on 1,509 participants, with the relative proportion of data types expected to meet or exceed those described above.

Data type	Core/ Requirement	Expected amount (existing / projected 2026-2031)	Additional details
UDS clinical and behavioral data	Clinical & DMS ¹ Cores / required by NACC ² UDS ³	1009/1509	Data kept raw, processed, derived and summarized; collected at baseline and annual follow-up (each visit); data to be shared via NACC and UF-secure encrypted repository (for limited data not broadly available)
Center-specific clinical and questionnaire data	Clinical & DMS Cores / required by Center	1009/1509	Data kept raw, processed, derived and summarized; collected at baseline and annual follow-up (each visit); data to be shared via NACC and UF-secure encrypted repository (for

			limited data not broadly available)
MRI	Clinical, Neuroimaging & DMS Cores / required by Center, SCAN ⁴ initiative	764/1096	Data kept raw, processed, and derived. MRI are acquired at the baseline visit for each enrolled participant; follow-up imaging is not currently within scope. Data are uploaded to LONI ⁵ via IDA ⁶ (SCAN Upload Portal ⁴)
Amyloid PET	Clinical, Neuroimaging & DMS Cores / required by Center, SCAN initiative	605/788	Raw, processed, and summarized data; collected for 24 ADRC and 22 CLARiTi ⁹ participants annually; Data are uploaded to LONI ⁵ via IDA ⁶ (SCAN Upload Portal ⁴)
Tau PET	Clinical, Neuroimaging & DMS Cores / required by Center, SCAN initiative	0/230	Raw, processed, and summarized data; collected for 24 ADRC and 22 CLARiTi ⁹ participants annually; Data are uploaded to LONI ⁵ via IDA ⁶ (SCAN Upload Portal ⁴)
ALZpath pTau-217 (fluid biomarker)	Clinical, Biomarker & DMS Cores / required by Center, NCRAD ⁷ ADCFB ⁸ Initiative	742/1242	Processed data; collected at baseline and at each time point (follow-up visit); data available via NACC Portal, ADCFB Data Visualization, and UF-secure encrypted repository (for limited data not broadly available)
Nf-L (fluid biomarker)	Clinical, Biomarker & DMS Cores / required by Center, NCRAD ADCFB Initiative	655/1,155	Processed data; collected at baseline and each time point (follow-up visit); data available via NACC Portal, ADCFB Data Visualization, and UF-secure encrypted repository (for limited data not broadly available).
GFAP (fluid biomarker)	Clinical, Biomarker & DMS Cores / required by Center, NCRAD ADCFB Initiative	656 /1,156	Processed data; collected at baseline and each time point (follow-up visit); data available via NACC Portal, ADCFB Data Visualization, and

			UF-secure encrypted repository (for limited data not broadly available)
NULISaseq CNS Disease 120 panel	Clinical, Biomarker & DMS Cores / required by Center,	0 /46	Raw, processed, and summarized data; collected for 24 ADRC and 22 CLARiTI ⁹ participants annually; data shared via UF-secure encrypted repository (for limited data not broadly available).
APOE genotype (genetic/genomic data)	Clinical, Biomarker & DMS Cores / required by Center, NACC, NCRAD ADCFB Initiative	716 / 1216	Processed data; collected at baseline and each time point; data available via NACC Portal and UF-secure encrypted repository (for limited data not broadly available).
Neuropathology or autopsy data	Clinical, Neuropathology & DMS Cores / required by Center, NACC	30 / unknown, contingent on future consents	Data kept raw, processed, derived and summarized; collected upon death of participant. Quantified data are entered into NACC's standardized Neuropathology Data Set (NP), and neuropathology data are uploaded through the ADRC Portal. Data available via NACC Portal and UF-secure encrypted repository (for limited data not broadly available).

¹ Data Management and Statistics

² National Alzheimer's Coordinating Center

³ Uniform Data Set

⁴ Standardized Centralized Alzheimer's & Related Dementias Neuroimaging, an NIA-funded initiative that standardizes MRI and PET acquisition, processing, quality assurance, and sharing across the Alzheimer's Disease Research Centers

⁵ Laboratory of Neuroimaging, a neuroinformatics center at the University of Southern California (USC)

⁶ Image and Data Archive, the secure repository/database operated by LONI

⁷ National Centralized Repository for Alzheimer's Disease and Related Dementias

⁸ Alzheimer's Disease Center Fluid Biomarker Initiative

⁹ Consortium for Clarity in Alzheimer's Disease and Related Dementias Research Through Imaging

Data include required elements of the National Alzheimer's Coordinating Center Uniform Data Set (UDS), which encompasses demographic, participant and family health history, physical and blood pressure evaluation, the Clinical Dementia Staging Instrument, measures of neuropsychiatric symptoms, geriatric depression, functional status, neurological examination, neuropsychological testing scores, and clinician diagnosis. The UDS also includes additional clinical and behavioral measures associated with Lewy Body Disease and Frontotemporal Dementia modules.

The UDS is supplemented with site-specific data collection, which includes neuroimaging data (structural MRI, diffusion tensor imaging, amyloid PET, tau PET; both raw images and post-processing quantifications are retained), tissue (Formalin-fixed paraffin embedded tissue block from 18 standardized brain regions, Formalin fixed brain tissue and frozen brain tissue) and fluid samples (plasma, serum and buffy coat for 715 participants), and fluid biomarkers (APOE Genotyping, A β 40/42, pTau-181 pTau-217, Nf-L, GFAP, neuroinflammatory cytokines, α Syn, comprehensive metabolic panel (CMP), a lipid panel, Hemaglobin A1c (HbA1c), creatinine, and C-reactive protein (CRP)).

Additional site-specific quantitative data include a modified Clinical Dementia Rating scale, a site-specific neuropsychological testing battery, as well as additional questionnaire-based measures of mild behavioral impairment, pain, sleep, acculturation, as well as language proficiency and experience.

Additional neuropathology and autopsy data (including semiquantitative analysis of beta-amyloid, tau, alpha-Synuclein and TDP-43 pathology, as well as assessment of cerebral amyloid angiopathy, vascular pathology and other proteinopathies) are available on a subset of participants.

SCIENTIFIC DATA THAT WILL BE PRESERVED AND SHARED, AND THE RATIONALE FOR DOING SO

UDS data are shared with NACC and are available through their public repository. Data are uploaded weekly, with a monthly inventory provided by NACC. NACC produces four quarterly frozen data sets annually, and shares these with all qualified users through their public use data portal.

We have shared over 700 samples with NCRAD to date. After completing the assays planned a priori for 1FL ADRC participants in this renewal (assays related to vascular, neurodegeneration, inflammation, and other vital clinical indicators), we will continue to share up to 3mL of plasma per participant with NCRAD.

For all quantified data, data are compiled into several deidentified, limited or comprehensive data sets. Access to these data proceeds through a four-step process: (a) submission of a data request to the 1Florida Executive Committee; (b) approved data requesters are credentialed through the University of Florida information security process; (c) credentialed requesters obtain either "quick access" comprehensive data sets (which may be deidentified or limited, according to user needs and IRB protocol), or custom data sets (restricted to particular measures, occasions, or participant types). Credentialing ensures that data can be shared through a HIPAA-compliant encrypted cloud storage portal that tracks data downloads and users.

Data are available in multiple formats for data analysis (i.e., csv, SAS, SPSS, Stata; longitudinal data sets may be provided in wide or long format).

Imaging and fluid specimen data are shared freely with 1FL ADRC investigators. Outside investigators can request samples and data through the 1FL ADRC website and this is managed by the DMS Core. In coordination with the DMS Core, requests are evaluated for sample availability and then submitted to the executive committee for evaluation and approval.

Metadata, other relevant data, and associated documentation: Briefly list the metadata, other relevant data, and any associated documentation (e.g., study protocols and data collection instruments) that will be made accessible to facilitate interpretation of the scientific data.

Metadata are stored for imaging and fluid biomarker data. Metadata are recorded and included in our comprehensive, quick access and custom data sets.

For MRI, diffusion tensor and amyloid pet and tau pet imaging, metadata include date of scanner, scanner type, tracer used (TR, TE, voxel size, orientation, magnetic field strength, #directions, b-values, flip angle, protocol name, data quality measures). For biomarker assays, metadata stored include dates of sample processing, experimentation, and data analysis, sample aliquot tested, test kits and lot numbers, and final quantification and QC values.

ELEMENT 2: RELATED TOOLS, SOFTWARE AND/OR CODE

Validation and quality control of UDS data occur during upload to the main NACC database maintained by the University of Washington. NIA-funded ADRCs upload data through password protected membership portals. Consistency and quality of UDS data is evaluated using proprietary SAS scripts developed and maintained by NACC staff. Data that does not pass validation must be corrected by ADRCs to be included in the dataset. Bulk submission data files can be reviewed by Centers for quality control outside the NACC framework using the open source NACCulator tool made available by the CTS-IT of the University of Florida on their GitHub site.

We have developed several tools for the processing of quantitative data and preparation for transmittal to NACC. These include

- RED-I Dropper (an open-source utility written using Python, JavaScript, and SQL which allows for secure transfer of data among research entities; it is customizable to restrict access to relevant investigators; code for this utility is available in GitHub).
- NACCulator, developed in Python, to convert REDCap's comma separated value (CSV) format into fixed width format for upload to the National Alzheimer's Coordinating Center's (NACC) submission systems enforcing NACC's coding rules; code for this utility is available in GitHub and has already been shared with and used by multiple ADRCs (e.g., University of Michigan).
- NACC Utilities creates a virtual machine (Docker) process to automatically upload data from NACCulator to the NACC website; this software has been shared with the Massachusetts ADRC to assist in their upload process of data from R scripts to NACC;
- DxSter is an open-source software developed to serve as an AD algorithmic diagnostic helper. The software is a codified algorithm that can assist in diagnosing normal cognition, MCI, and dementia.
- NACCuum Cleaner, developed in R, pulls data from REDCap through API and combines it with all other center data into a comprehensive dataset that is readily analyzable. In the process of combining, data from each data source undergo numerous quality checks and are cleaned and restructured as necessary.

Regarding software used to process data:

RStudio is a free R development environment that runs on most operating systems. R Scripts produced through the course of the research will be made publicly available on the lab's GitHub repository and will be provided as Supplementary files for any publications through a Zenodo-GitHub link. R is a free software environment for statistical computing and graphics.

For individual manuscripts, code will be available no later than when a publication has been submitted.

Some investigators conduct analyses in SAS, SPSS, MPlus and Stata. These packages are available for purchases, and many are available at reduced cost through partner academic institutions at the University of Florida and University of Miami.

Regarding software used to process images:

FreeSurfer software is used for volumetric analysis and is available from (<http://surfer.nmr.mgh.harvard.edu/>). FMRIB Software Library (FSL) is used for diffusion imaging analysis and is available from (<https://fsl.fmrib.ox.ac.uk/fsl/fslwiki/>). AFNI (<https://afni.nimh.nih.gov/>) is used for creating imaging pipeline and extracting meta data. ANTs (<https://github.com/ANTsX/ANTs>) and ANTsR (<https://github.com/ANTsX/ANTsR>)

are used for imaging normalization. MATLAB (MathWorks, MA) is used for free-water imaging analysis and is available through the University of Florida.

ELEMENT 3: STANDARDS

Data dictionaries and metadata level standardization of the described NACC data types are freely available and outlined in the “Forms & documentation” section of the NACC website (DOI:10.6069/0wbkxvqg). Conventions around variable naming, data quality, and continuity are provided by NACC and described in these data dictionaries for all data types described in Element 1.

MRI and PET data for ADRC participants will be collected in accordance with the acquisition protocols outlined by Standardized Centralized Alzheimer’s and Related Dementias Neuroimaging (SCAN). Non-SCAN compliant MRI and PET images and biomarker data are available upon request under executed DUA. Training materials related to NACC UDS administration are also made available online. “Coding Guidebooks” and the “Instructions for the Neuropsychology Battery” contain NACC guidelines for instrument administration that may differ from guidelines in instrument specific documentation. Examples include the MoCA, classic CDR interview administration, NPI-Q administration, and the FTL module additions to the CDR.

In accordance with FAIR Principles for data, we will use open file formats (e.g. JPEG, MP4, CSV, TXT, PDF, HTML, etc.) and persistent unique identifiers (PIDs) such as NACCIDs for resources (e.g., antibodies, cell lines, software tools, and databases) and DOIs for protocols using protocols.io. The bioimaging community has not yet agreed on a single standard data format that is generated by all acquisition systems, but we will use OME-Files for data that will be preserved and shared. Beginning with this renewal, we will work with NIA and NACC to implement globally unique identifiers (GUID).

ELEMENT 4: DATA PRESERVATION, ACCESS, AND ASSOCIATED TIMELINES

Data Type	Where Deposited?	When Deposited?	How and when shared with qualified investigators?	How long will data be available?
UDS clinical and behavioral data	NACC and encrypted secure institutional storage	Continuously upon acquisition and processing	(1) Via NACC; (2) Via 1Florida Executive Committee request; on demand, securely encrypted transfer methods that are institutionally approved	Indefinitely. Preserved in perpetuity under NIA-supported NACC archival infrastructure
Center-specific clinical and questionnaire data	Encrypted secure institutional storage. Each monthly freeze is assigned a version identifier and retained	Continuously upon acquisition and processing	Via 1Florida Executive Committee request; on demand, securely encrypted transfer methods that are	Availability for sharing indefinitely, based on availability of funding. Data for specific manuscripts may also be available with journal repositories.

			institutionally approved	
Human MRI (raw, SCAN-complaint)	LONI IDA (SCAN upload portal) Scans are indexed by LONIUID at upload and mapped to NACCID, providing persistent unique identifiers across the three 1Florida sites	Continuously upon acquisition and processing	(1) Via NACC and SCAN; (2) Via 1Florida Executive Committee request; on demand, securely encrypted transfer methods that are institutionally approved	Indefinitely. Preserved in perpetuity under NIA-supported LONI/NACC archival infrastructure
Human amyloid PET (raw, SCAN-compliant)	LONI IDA (SCAN upload portal) Scans are indexed by LONIUID at upload and mapped to NACCID, providing persistent unique identifiers across the three 1Florida sites	Continuously upon acquisition and processing	(1) Via NACC and SCAN; (2) Via 1Florida Executive Committee request; on demand, securely encrypted transfer methods that are institutionally approved	Indefinitely. Preserved in perpetuity under NIA-supported LONI/NACC archival infrastructure
Human Tau PET (raw, SCAN-compliant)	LONI IDA (SCAN upload portal) Scans are indexed by LONIUID at upload and mapped to NACCID, providing persistent unique identifiers across the three 1Florida sites	Continuously upon acquisition and processing	(1) Via NACC and SCAN; (2) Via 1Florida Executive Committee request; on demand, securely encrypted transfer methods that are institutionally approved	Indefinitely. Preserved in perpetuity under NIA-supported LONI/NACC archival infrastructure
Non-SCAN imaging (local sequences)	Encrypted secure institutional storage	Continuously upon acquisition and processing	Via 1Florida Executive Committee request; on demand, securely	Availability for sharing indefinitely, based on availability of funding.

			encrypted transfer methods that are institutionally approved	
Derived imaging measures (FreeSurfer volumetrics, DTI scalars, PET SUVRs, QC metrics)	Encrypted secure institutional storage. Each monthly freeze is assigned a version identifier and retained	Continuously upon acquisition and processing	Via 1Florida Executive Committee request; on demand, securely encrypted transfer methods that are institutionally approved	Availability for sharing indefinitely, based on availability of funding. Data for specific manuscripts may also be available with journal repositories.
ALZpath pTau-217 (fluid biomarker)	NACC ADRC Portal	Continuously upon acquisition and processing	(1) Via NACC; (2) Via 1Florida Executive Committee request; on demand, securely encrypted transfer methods that are institutionally approved	Indefinitely. Preserved in perpetuity under NIA-supported NACC archival infrastructure.
Nf-L (fluid biomarker)	NACC ADRC Portal	Continuously upon acquisition and processing	(1) Via NACC; (2) Via 1Florida Executive Committee request; on demand, securely encrypted transfer methods that are institutionally approved	Indefinitely. Preserved in perpetuity under NIA-supported NACC archival infrastructure.
GFAP (fluid biomarker)	NACC ADRC Portal	Continuously upon acquisition and processing	(1) Via NACC; (2) Via 1Florida Executive Committee request; on demand, securely encrypted transfer methods that	Indefinitely. Preserved in perpetuity under NIA-supported NACC archival infrastructure.

			are institutionally approved	
<i>APOE</i> genotype (genetic/genomic data)	NACC ADRC Portal	Continuously upon acquisition and processing	(1) Via NACC; (2) Via 1Florida Executive Committee request; on demand, securely encrypted transfer methods that are institutionally approved	Indefinitely. Preserved in perpetuity under NIA-supported NACC archival infrastructure.
Neuropathology or autopsy data	NACC ADRC Portal	Continuously upon acquisition and processing	(1) Via NACC; (2) Via 1Florida Executive Committee request; on demand, securely encrypted transfer methods that are institutionally approved	Indefinitely. Preserved in perpetuity under NIA-supported NACC archival infrastructure.

As a general statement, **scientific data will be shared no later than the time of an associated publication or the end of the performance period, whichever comes first.**

ADRCs will utilize NACC as the data repository for UDS data collected from ADRC participants. Data housed in the NACC system includes ADRC site identifiers and participant identifiers unique for each participant. UDS data and metadata for all previously described data types are preserved through routine uploads of validated data to the NACC database systems. Preservation of other data types are conducted by NCRAD (biological samples), SCAN (raw neuroimaging data), and NIAGADS (genotypic data) with access for all data facilitated by NACC. Data types are aligned by mapping of site-specific participant IDs to NACC IDs using the “PTID to NACCID Map” in the ADRC Portal.

Other data requesters accept the terms of the NACC Data Use Agreement and can submit requests for data for formal review at the NACC website. Data request policies and expectations are outlined in the NACC Handbook UDSv4. Publications utilizing NACC data are required to acknowledge NACC grants and submit abstracts and manuscripts to NACC for review during the publication process.

ADRCs upload NACC data at regular intervals and its associated collaborators according to data types and metadata listed above. NACC performs quality control and derivation of variables requiring calculations on new data added to the repository on a quarterly basis referred to as a data freeze. Each data freeze is given a unique identifier to allow easier comparisons of data analysis.

Site-specific data are stored as de-identified and limited data sets on an encrypted cloud-based storage portal licensed by University of Florida (via secure encrypted institutionally approved storage.). To download these

data, users undergo a quick University of Florida credentialing process (which includes information security and HIPAA training), and then receive credentials to download data from the encrypted portal. For each data release, we track the investigator's contact information, IRB protocol and approval.

All raw MRI and PET images are de-identified and are preserved in locked buildings and locked data centers. ,SCAN-compliant scans (MRIs and PETs) are transferred to SCAN via LONI IDA. Non-SCAN imaging data are shared upon request under DUA. Derived imaging measures are stored in a UF-approved data storage application, with quarterly frozen versions maintained for data tracking and reproducibility. The National Centralized Repository for Alzheimer's Disease and Related Dementias (NCRAD) banks and distributes biospecimens to qualified investigators. The NIA Genetics of Alzheimer's Disease Data Storage Site (NIAGADS) serves as the NIA-designated repository for AD/ADRD genetic data where controlled access datasets are released to qualified investigators. The Alzheimer's Disease Genetics Consortium (ADGC) generates genome-wide array and sequence data on ADRC-derived cohorts ascertained through NACC and NCRAD; data is distributed through NIAGADS.

Per July 2026 numbers provided by NACC, NACC provides longitudinal clinical data on more than 50,000 participants through its Uniform Data Set (UDS; 2005-present), with over 200,000 study visits available for research. In addition, the Minimum Data Set (MDS), offers abstracted data on ADRC subjects from 1984 to 2005. A large number of these also have neuropathology data. NACC, NCRAD, NIAGADS, and the Alzheimer's Disease Research Centers (ADRCs) work together to track phenotypic data, biologic specimens, and genotypic data from ADRC participants. Researchers can request data or specimens from the relevant NIA-supported resource. Often, multiple types of data are available for the same participants — in which case, participant identifiers can be used to tie the various available data together.

NCRAD's catalogs of biospecimens include a limited data set from the NACC uniform data set (UDS). In addition, the catalogs provide links to NIAGADS datasets for subjects with available genetic data. NCRAD is the NIA-designated biorepository for biospecimens from dementia-related studies. It banks a broad range of samples that can be requested by qualified investigators. These include DNA, RNA, plasma, serum, CSF, brain tissue, lymphoblastoid cell lines (LCLS), peripheral blood mononuclear cells (PBMCS), induced pluripotent stem cells (IPSCS), fibroblasts, and feces. The scientific data will be findable and identifiable via a persistent unique identifier or other standard indexing tools (GUID):

We will use Persistent Unique Identifiers (PIDs) to improve data findability across all dissemination outputs. PIDs used will include ORCID iDs and NIH user IDs for people, DOIs for outputs (e.g., datasets, protocols), Research Resource Identifiers (RRIDs) for resources, and Research Organization Registry (ROR) IDs and funder IDs for places, as much as possible to make data identifiable and findable. We will also use indexed metadata, such as MeSH terms with a unique URL to make scientific data easily findable. We will keep our ORCID and NIHMS records up to date with DOIs for our datasets and publications, ROR, and funder IDs to increase findability.

WHEN AND HOW LONG THE SCIENTIFIC DATA WILL BE MADE AVAILABLE

All scientific data generated from this project will be made available as soon as possible and will appear in regular data uploads to NACC (for UDS data elements). Site-specific data are updated daily in real time, and integration of UDS and 1Florida data occurs immediately after each quarterly NACC data freeze. Sharing of UDS data will be perpetual, contingent on the maintenance of NACC data resources. Site-specific 1Florida data will be available indefinitely based on the availability of funding, including MRIs, PETs, blood samples, and skin biopsies. **As a core rule: all scientific data will be shared no later than the time of an associated publication or the end of the performance period, whichever comes first.**

ELEMENT 5: ACCESS, DISTRIBUTION, OR REUSE CONSIDERATIONS

Factors affecting subsequent access, distribution, or reuse of scientific data: NIH expects that in drafting Plans, researchers maximize the appropriate sharing of scientific data. Describe and justify any applicable factors or data use limitations affecting subsequent access, distribution, or reuse of scientific data related to informed consent, privacy and confidentiality protections, and any other considerations that may limit the extent of data sharing.

To address safety and security concerns related to sharing data regarding clinical health and cognitive data, we take the following precautions: (1) all data collection is subject to an sIRB administered at the University of Florida; this sIRB carefully monitors all questionnaires and measures, procedures and data dissemination plans; (2) all investigators are asked to provide IRB approval for their research uses of data; for uses classified to be exempt from IRB oversight (e.g., non-human, deidentified data), we direct users to quick-determination tools that certify their research as exempt; (3) we credential all users prior to receipt of data through a University of Florida credentialing system that ensures users have information security and HIPAA training; (4) we track all data disclosures and report disclosures to the IRB as required; (5) unless specific research needs necessitate otherwise, all data shared include no defined Personal Health Information elements including dates (instead, dates are converted to ages or days since baseline testing); this is referred to as the "Safe Harbor" method of data deidentification.

All imaging scans collected at three clinical sites are uploaded to a cloud server hosted at the University of Florida through the 1FL ADRC Data Transfer Portal. This is secure web-based software that allows large data storage and dissemination. Sensitive data and protected health information are stripped away and de-identified at the user's browser side before scans are uploaded to the server. This technology has been approved at each clinical site and was originally developed in U01 NS119562 and then adapted for the 1FL ADRC. The software implements dual-factor authentication login and is encrypted with a Secure Connection Protocol approved by the UF Integrated Risk Management system. The functions of the transfer tool have improved significantly on security and processing speed through the current funding cycle and are able to accommodate a variety of networking speeds and systems with different computational capacities.

Whether access to scientific data will be controlled: State whether access to the scientific data will be controlled (i.e., made available by a data repository only after approval).

ADRC site level consent development is supported through NACC resources regarding consent best practices, which include language around data storage, access, and capacity assessment: (DOI:10.6069/p6jw2zxf). Other data requesters accept the terms of the NACC Data Use Agreement (DUA) and can submit requests for data for formal review at the NACC website. Data request policies and expectations are outlined in the NACC Handbook (DOI:10.6069/g8v9cdx7). Publications utilizing data uploaded to NACC are required to acknowledge NACC grants and submit abstracts and manuscripts to NACC for review during the publication process as outlined in the Authors Checklist found on the NACC website. Access to scientific data is made available through the NACC data request process and available after the signing of a DUA. Storing and sharing procedures through NACC systems are designed according to NIH GCDMP practices and standards which are outlined on the NACC website and the terms of the Data Use Agreement. All data users are required to complete and sign a NACC DUA prior to dataset access. The NACC Data Use Agreements are designed to ensure appropriate use of the data solely by the individuals identified in the data request, ensure NACC is informed of intended manuscript submission, and ensure that NACC (and all ADRCs submitting data to NACC) are appropriately acknowledged in any publications. All data users are required to complete and sign a Data Use Agreement prior to dataset access. All collaborators on a project requesting NACC data are bound by the terms of the DUA and can freely share provided datasets amongst each other. NACC data utilization is limited to the data streams outlined above. NACC distributes deidentified datasets to protect the privacy of study participants. All Data Requests are reviewed by the NACC Data Use Committee prior to distribution. The NACC Data Use Committee determines if researchers are provided with the Investigator or Commercial data file. This review ensures that participant's consent to participate in commercial-based research activities is respected.

To be in compliance with the University of Florida IRB, site-specific data beyond those shared with NACC (and which are available in NACC's public use repository) will be available by controlled access only as documented above. To get access to the data, the user must submit a valid scientific question, receive approval from our 1Florida ADRC Executive Committee, and provide IRB information. Users who seek to contact research participants for further research must have their contact registry requests approved by the 1Florida Executive Committee. Once approved, contact information is disclosed, and users are asked to provide final disposition of all contacted participants (e.g., contacted, enrolled, refused, unable to contact), and this is logged.

PROTECTIONS FOR PRIVACY, RIGHTS, AND CONFIDENTIALITY OF HUMAN RESEARCH PARTICIPANTS: IF GENERATING SCIENTIFIC DATA DERIVED FROM HUMANS, DESCRIBE HOW THE PRIVACY, RIGHTS, AND CONFIDENTIALITY OF HUMAN RESEARCH PARTICIPANTS WILL BE PROTECTED (E.G., THROUGH DE-IDENTIFICATION, CERTIFICATES OF CONFIDENTIALITY, AND OTHER PROTECTIVE MEASURES).

In a memo to investigators from the University of Florida Institutional Review Boards, Dated August 2024, the IRB stated that "**Certificates [of Confidentiality]** will be issued automatically to recipients for NIH-funded research that meets the scope of the NIH Policy. Certificates may be issued for non-federally funded research upon request but are not mandatory. Information protected by a Certificate and all copies are subject to the protections afforded by the Certificate in perpetuity. Therefore, if a secondary researcher receives information protected by a Certificate the secondary researcher is required to uphold the protections afforded by the Certificates. NIH expects that recipients of a Certificate will inform secondary researchers when information disclosed to them is protected by a Certificate."

To ensure participant consent for data sharing, IRB paperwork and informed consent documents will include language describing plans for data management and sharing of data, describing the motivation for sharing, and explaining that personal identifying information will be removed. To protect participant privacy and confidentiality, shared data will be de-identified using the Safe Harbor method, which removes the following data elements prior to sharing:

- Names
- All geographic subdivisions smaller than a state except for the initial three digits of the ZIP code
- All elements of dates
- Telephone numbers
- Vehicle identifiers and serial numbers, including license plate numbers
- Fax numbers
- Device identifiers and serial numbers
- Email addresses
- Web Universal Resource Locators (URLs)
- Social security numbers
- Internet Protocol (IP) addresses
- Medical record numbers
- Biometric identifiers, including finger and voice prints
- Health plan beneficiary numbers
- Full-face photographs and any comparable images
- Account numbers
- Any other unique identifying number, characteristic, or code
- Certificate/license numbers

ELEMENT 6: OVERSIGHT OF DATA MANAGEMENT AND SHARING

DESCRIBE HOW COMPLIANCE WITH THIS PLAN WILL BE MONITORED AND MANAGED, FREQUENCY OF OVERSIGHT, AND BY WHOM AT YOUR INSTITUTION (E.G., TITLES, ROLES).

Lead MPIs Glenn E. Smith (0000-0003-1506-9484) AND David A. Loewenstein (0000-0001-6788-2798) will be responsible for the day-to-day oversight of lab/team data management activities and data sharing, for which day-to-day oversight is delegated to Data Core Leader Michael Marsiske (0000-0001-5973-2116), Data Manager Christopher Barnes, and Project Manager Christy Rodkin. Broader issues of DMS Plan compliance oversight and reporting will be handled by the PI and Co-I team as part of general stewardship, reporting, and compliance processes. Annual progress reports (including minor deviations) are made to the sIRB of record at University of Florida (as well as real-time major deviations and protocol revisions). In addition, an internal and external advisory board, the latter of which includes representatives of the funding agency, receive regular progress reports. ADRCs can monitor their NACC and UDS performance, data quality, and data summarizations through monthly reports from NACC, which are also sent to NIA. ADRC member sites have specific membership credentials allowing access to their working datasets for uploading of site NACC data and confirming accuracy.

Compliance-monitoring is done at several levels and frequencies:

- *Ongoing*: Tracking of data management system is restricted to credentialed users, and all access is logged and tracked
- *Weekly*: DMS staff meet with data management staff at each data collection site at least weekly, and reports are communicated to leadership in real time
- *Biweekly*: The DMS team meets every other week, and reviews compliance with all DMS plan elements. Meetings are logged with minutes via a secure in-house channel. Problems are communicated immediately to the PIs and Project Manager.
- *Twice-monthly*: The Executive Committee meets at least twice a month. Protocol violations, including deviation from the DMS plan, are communicated as a standing item on that agenda.
- *Annually and ongoing*: At least once a year there is a meeting of the 1Florida ADRC External Advisory Committee. Summaries of protocol and data management compliance are provided at these meetings to help inform the EAC report. Serious breaches would be reported in real time.

Responding to problems or changes in the approved Plan:

Isolated logistical problems with the DMSP will be addressed as each arises to ensure smooth operation of data requests, data provisioning, and data sharing. Broader, systemic problems with implementation of the DMSP will be addressed at the bimonthly Executive Committee meeting.

Should an incident arise where data is released inconsistent with University of Florida data sharing policies, the appropriate offices (IRB, privacy office, etc) will be notified per university policy. Actions to remedy the incident will be taken in a timely manner, the incident will be reviewed at the bimonthly Executive Meeting, and corrective actions will be taken as needed.

Changes, corrections, or updates made to the DMSP during the year, will be included in the Center's annual RPPR per NIH policy.

The data management plan is a part of the IRB-approved SIRB protocol for this study. Any changes would need to be pre-approved by the University of Florida IRB prior to implementation. Each clinical core site completes a monthly reporting log including information regarding consents, adverse events, and deviations and these are shared with the regulatory contact at the University of Florida. The regulatory contact reviews the logs and maintains a list of minor deviations which are reported annually at the time of the annual IRB review. Major protocol violations are reported to the SIRB when they are identified through communications, the monthly reporting logs, or other means. As noted above, adverse events and protocol deviations are standing agenda items at the executive committee meetings which occur twice/month. Data are released only through the Data, Neuropathology, and Biomarker cores, not through individual sites. Any problems identified by those cases will be shared with the regulatory contact/SIRB team.

PLANNED RESEARCH OUTPUTS

DATASET - "UNIFORM DATA SET"

Clinical, behavioral and survey data collection including demographic, participant and family health history, physical and blood pressure evaluation, the Clinical Dementia Staging Instrument, measures of neuropsychiatric symptoms, geriatric depression, functional status, neurological examination, neuropsychological testing scores, and clinician diagnosis.

PLANNED RESEARCH OUTPUT DETAILS

Title	Type	Anticipated release date	Initial access level	Intended repository(ies)	Anticipated file size	License	Metadata standard(s)	May contain sensitive data?	May contain PII?
Uniform Data Set	Dataset	Unspecified	Open	NACC		None specified	NACC Uniform Data Set	Yes	No
Tissues / fluids	Dataset	unspecified	Open	NACC		None specified	NACC Brief Data set and Neuropathology Data set	Yes	No
SCAN	Images	Unknown	Qualified researchers	NACC/ SCAN	1G per sample	None specified	NACC/ SCAN	Yes	No